

1. What is the difference between **WHERE** and **HAVING** in SQL?

Answer:

- **WHERE** filters rows **before grouping**
- **HAVING** filters groups **after aggregation**

Example:

```
SELECT department, COUNT(*)  
FROM employees  
WHERE salary > 30000  
GROUP BY department  
HAVING COUNT(*) > 5;
```

Explanation:

- **WHERE** removes employees with salary ≤ 30000
 - **HAVING** keeps only departments having more than 5 employees
-

2. What is the difference between **INNER JOIN** and **LEFT JOIN**?

Answer:

INNER JOIN

Returns only matching rows from both tables.

LEFT JOIN

Returns all rows from the left table and matching rows from the right table.

Example:

```
SELECT c.customer_name, o.order_id  
FROM customers c  
LEFT JOIN orders o  
ON c.customer_id = o.customer_id;
```

Explanation:

Even customers without orders will appear in **LEFT JOIN**.

3. What are NULL values in SQL?

Answer:

NULL means missing or unknown data.

It is not:

- zero
- blank space
- false

Important:

You cannot use **=** with NULL.

 Wrong:

WHERE salary = NULL

 Correct:

WHERE salary IS NULL

4. What is data cleaning in data analysis?

Answer:

Data cleaning means preparing raw data before analysis.

It includes:

- removing duplicates
- handling missing values
- fixing inconsistent formats
- correcting errors

Example:

Changing:

- **Male, male, M**

into a single standardized value:

- Male

Why important?

Bad data leads to wrong insights and poor business decisions.

5. Explain the steps of a data analysis project.

Answer:

Typical workflow:

1. Understand business problem
2. Collect data
3. Clean data
4. Analyze data
5. Visualize insights
6. Share recommendations

Example:

For an e-commerce company:

- analyze sales trends
- identify top-selling products
- find customer behavior patterns
- recommend strategies to increase revenue